

Math 2 • Replacements review

Only the nine Math 2 slots replaced or inserted (A4, B9, B23, C17, C18, C24, D17, D23, E1). For checking - not a full paper.

A4 • REPLACE

MM7.5 trapezium rule (was trig integration)

The table gives values of a function f .

x	0	1	2	3
$f(x)$	2	3	6	11

It is also known that $f''(x) > 0$ for $0 < x < 3$. The trapezium rule with these four ordinates is used to estimate

$$\int_0^3 f(x) dx.$$

Which statement is correct?

- A 15, underestimate
- B 15, overestimate
- C $\frac{31}{2}$, underestimate
- D $\frac{31}{2}$, overestimate
- E 16, underestimate
- F 16, overestimate

B9 • REPLACE

MM4.1 ambiguous sine rule

In triangle ABC ,

$$\angle A = 30^\circ, \quad BC = 4, \quad CA = 4\sqrt{3}.$$

What is the sum of the areas of all possible non-congruent triangles ABC satisfying these conditions?

- A $4\sqrt{3}$
- B $8\sqrt{3}$
- C $12\sqrt{3}$
- D $16\sqrt{3}$
- E $20\sqrt{3}$

B23 • REPLACE

MM8.2 transformations (cut S_inf repetition)

The graph of $y = f(x)$ has exactly three x -intercepts:

$$-4, \quad 2, \quad 8.$$

A new function is defined by

$$g(x) = 3f(2x - 4).$$

What is the sum of the x -coordinates of the x -intercepts of $y = g(x)$?

- A** 1
- B** 3
- C** 5
- D** 7
- E** 9
- F** 11

C17 • INSERT

fill missing Q17; MM6 stationary-point reasoning

A cubic polynomial f satisfies

$$f'(x) = 3(x - 1)(x - 3).$$

Which statement must be true?

- A** $f(1) > f(3)$
- B** $f(1) < f(3)$
- C** $f(1) = f(3)$
- D** f has a local minimum at $x = 1$
- E** f has no local maximum
- F** None of the above

C18 • REPLACE

MM2.3 finite GP (was broken)

The first five terms of a geometric progression are all positive. The sum of the first three terms is 7. The sum of the final three terms is 28. What is the sum of all five terms?

- A** 21
- B** 28
- C** 31
- D** 35
- E** 42
- F** 49

C24 • REPLACE

MM7.6 $dy/dx = f(x)$ polynomial DE

A curve satisfies

$$\frac{dy}{dx} = 6x^2 - 30x + 24.$$

The curve has two stationary points. What is the vertical distance between these stationary points?

- A** 9
- B** 18
- C** 21
- D** 24
- E** 27
- F** 30

D17 • REPLACE

MM2.1 recurrence (was log sum)

A sequence is defined by

$$u_1 = 2, \quad u_{n+1} = \frac{1}{1 - u_n}.$$

What is the value of u_{2026} ?

- A** -2
- B** -1
- C** $-\frac{1}{2}$
- D** $\frac{1}{2}$
- E** 2

D23 • REPLACE

MM8.2 function composition

Functions f and g are defined by

$$f(x) = x^2 - 1, \quad g(x) = 2x + 1.$$

The equation

$$f(g(x)) = g(f(x))$$

has two real solutions p and q . What is the value of $p + q$?

- A** -4
- B** -2
- C** -1
- D** 1
- E** 2

E1 • REPLACE

MM6.3 normal to a curve (was In integral)

The point P on the curve

$$y = x^3 - 3x$$

has x -coordinate 2. The tangent at P meets the y -axis at R , and the normal at P meets the x -axis at Q . What is the area of triangle OQR , where O is the origin?

- A** 80
- B** 128
- C** 144
- D** 160
- E** 180
- F** 320